Humanizing *Saccharomyces cerevisiae*:
Functional Rescue of Rare Disease Genes for Variant Effect Mapping

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Background

Many rare genetic diseases are caused by **missense mutations** which alter protein structure and subsequently function.

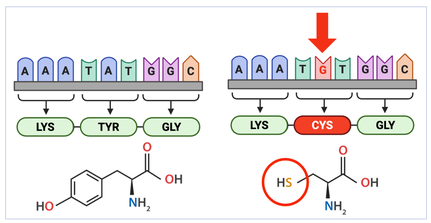
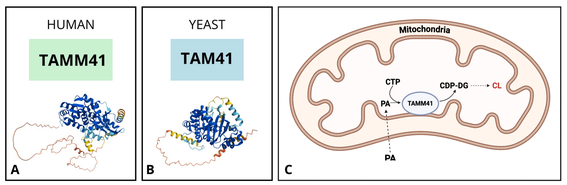


Figure 1. *Missense mutation Y110C in TAMM41 is 1 of 2 known pathogenic variants causing COXPD, as recorded on ClinVar.*

Combined Oxidative Phosphorylation Deficiency (COXPD) is a rare disease from TAMM41 mutations that impair mitochondrial protein synthesis, causing myopathy, hypotonia, and developmental delay.[1]



Mitochondrial phosphatidate cytidylyltransferase

For cardiolipin biosynthesis, a key phospholipid of the inner mitochondrial membrane

Figure 2. *TAMM41 homologues and function: (A) human, (B) yeast structures (AlphaFold); (C) cardiolipin biosynthesis.*

Though individually rare, these diseases collectively affect millions worldwide (~1 in 100), highlighting a broader challenge: **Limited data and many unknown variants.**

Our yeast-based platform enables **high-throughput functional mapping** to anticipate variant impact before it appears in patients.[2]

Results

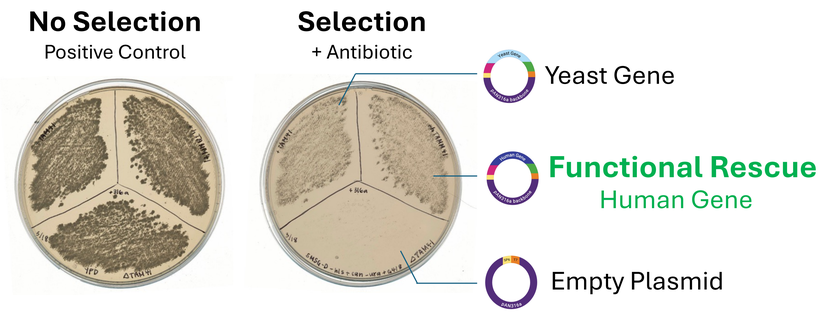
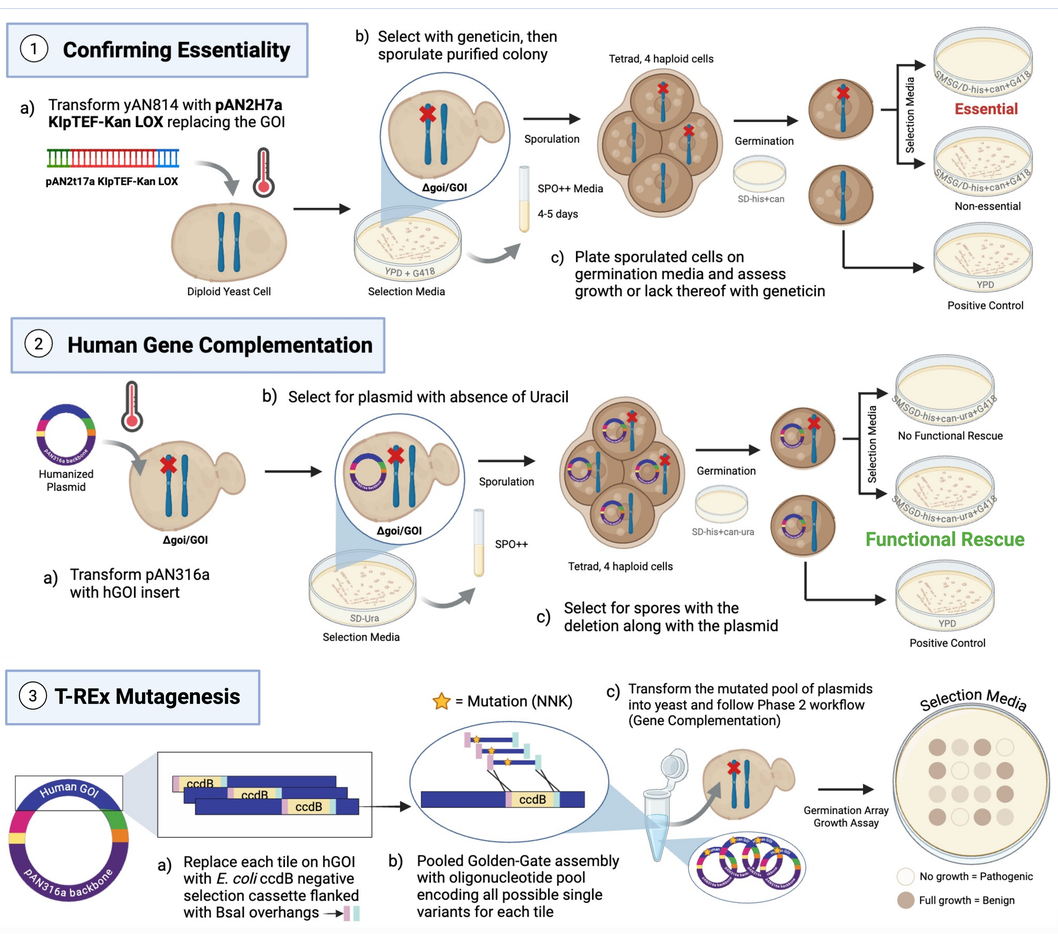


Figure 3. *Complementation of human TAMM41 in yeast enables functional assessment of missense variants, with controls confirming assay reliability.*

Methodology



Conclusions & Future Directions

Tiled-Region Exchange (T-REx) mutagenesis of TAMM41 is ongoing, with known pathogenic controls (Y110C, P137L) included to validate assay performance and enable visual assessment of functional readouts.[2] This workflow is currently extended to five additional genes.

The resulting experimentally generated **Variant Effect Map** will provide a functional framework for interpreting genetic variation, supporting precision diagnostics and predictive assessment of rare disease mutations.

AlphaFold Generated Variant Effect Map

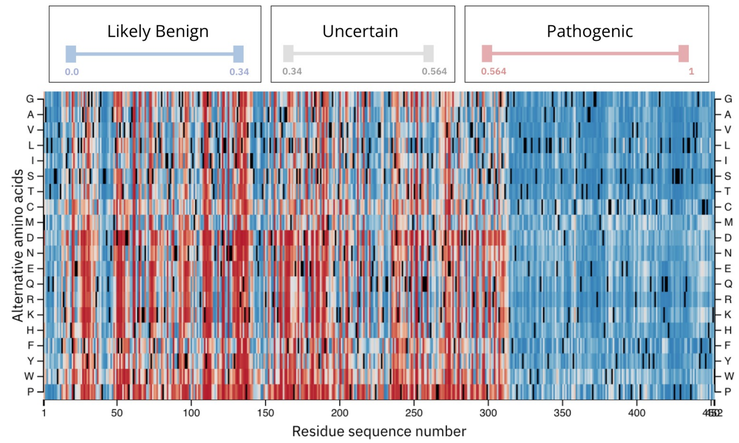


Figure 4. *Predicted TAMM41 variant effect map from AlphaFold models. Trained on evolutionary data, these predictions have limited accuracy for functional effects. Experimental validation is necessary.*

Acknowledgements

1. Thompson, K. *et al.* Biallelic variants in TAMM41 are associated with low muscle cardiolipin levels, leading to neonatal mitochondrial disease. *Human Genetics and Genomics Advances* **3**, 100097 (2022).
2. Kindree, K. *et al.* Deep-mutational scanning libraries using Tiled-Region Exchange mutagenesis. *G3: Genes, Genomes, Genetics* **16**, (2026).

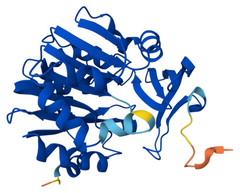
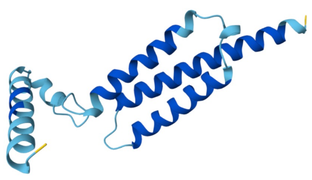
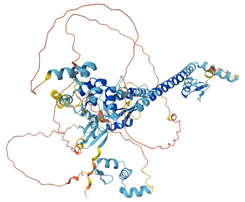
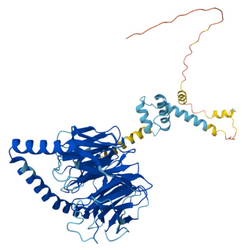
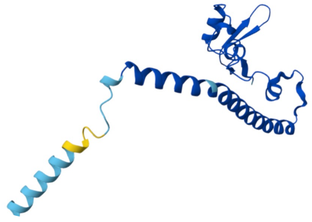
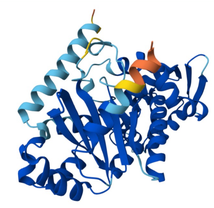
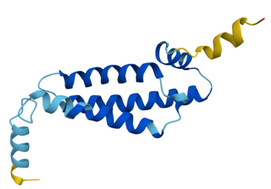
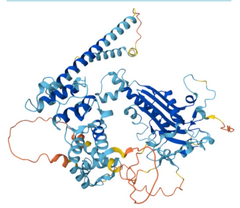
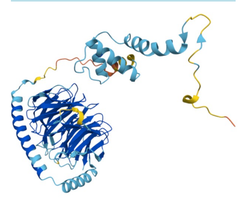
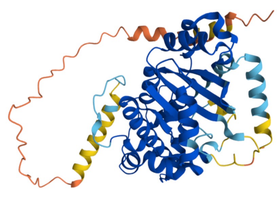
SGD Saccharomyces Genome Database
ALLIANCE of GENOME RESOURCES
NCBI National Center for Biotechnology Information
ClinVar Clinically relevant variation
AlphaFold
UniProt
Google DeepMind
bio RENDER
NIH National Library of Medicine National Center for Biotechnology Information
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Humanized *Saccharomyces cerevisiae*: Summary of Genes

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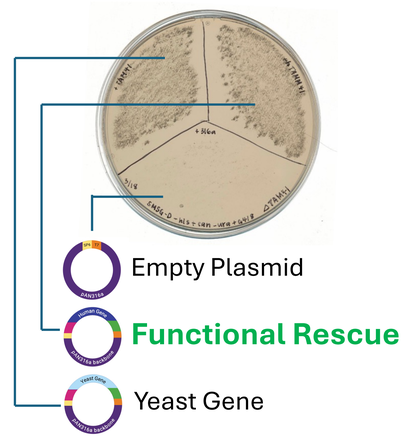
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HUMAN GENE	**SRM**	**MPC2**	**ANKZF1**	**PRPF4**	**RSL24D1**	**TAMM41**
						
PATHOGENIC ASSOCIATION	Snyder-Robinson Syndrome (X-linked disability)	Mitochondrial Pyruvate Carrier Deficiency (Neurological disorders)	Infantile-Onset Inflammatory Bowel Disease	Retinitis Pigmentosa	Diamond-Blackfan Anemia	Combined Oxidative Phosphorylation Deficiency (COXPD)
YEAST GENE	**SPE3**	**MPC2**	**VMS1**	**PRP4**	**RLP24**	**TAM41**
						
SHARED FUNCTION	Spermidine synthase (Polyamine biosynthesis)	Highly conserved subunit of the mitochondrial pyruvate carrier	Peptidyl-tRNA hydrolase; releases stalled peptides from ribosomes	Splicing factor; component of the U4/U6-U5 snRNP complex	Essential protein required for ribosomal large subunit biogenesis	Mitochondrial phosphatidate cytidylyltransferase

No Selection
Positive Control

Selection
+ Antibiotic





These human homologues did not achieve functional complementation in yeast; ongoing work focuses on optimizing conditions and construct design to enable rescue, and ultimately, generate variant effect maps.